

MyDesign Portfolio

Element I Initial Template

Test 1 – Ability to lay motionless on multiple different flat surfaces (Material differences)

Requirement Priority: High

(Requirement: The prototype must be able to remain motionless when laid on a flat surface for half a minute when not held by a person or object. The surface material will vary depending on the test)

Testing Procedure:

Independent Variable: Material that the flat surface is made of

Dependent Variable: If the prototype remains motionless

Controls: Angle of the surface (180 degrees), time required for completion (30 seconds)

Each test was repeated 3 times per material change position

All other conditions held constant

Raw Data:

Material	Time prototype is motionless (seconds)	Success Rate (between 3 trials)
Wood	30	100%
Metal	30	100%
Rubber	30	100%
Plastic	30	100%
Ceramic	30	100%

Graph 1: **N/A**

Photo 1: Metal



Photo 2: Wood



Photo 3: Rubber

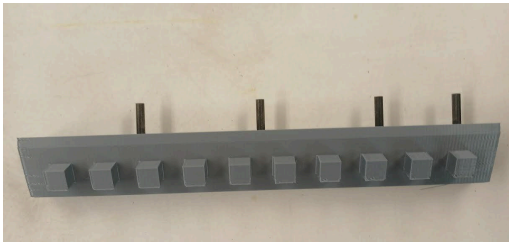


Photo 4: Plastic

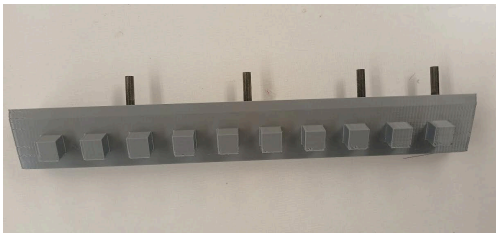


Photo 5: Ceramic



Analysis and Summary:

Based on our testing data, it appears that the prototype remains motionless when laid on a flat and level surface of all of the materials tested with 100% certainty. This is the desired outcome for the final product as workers in the shop will lay these down

Implications and Next Steps:

Our design is currently effective in this area, we must not compromise the prototypes stability in future iterations.

Test 2 – Ability to remain mounted on a wall with the weight of 10 ½ inch sockets

Requirement Priority: High

(Requirement: The prototype must remain mounted securely on a wall when it is bearing the weight of 10 ½ inch sockets (or above that size) for above 15 minutes)

Testing Procedure:

Independent Variable: Sockets on the object (½ only)

Dependent Variable: Time the object can remain on the wall

Controls: Pegboard material, no interference, weight and size of sockets and the prototype used, and height the object is placed at on the pegboard

The test was repeated 3 times

All other conditions held constant

Raw Data:

Trial #	Socket Size (In)	Socket amnt	Time on wall (sec)
1	>1/2	10	900
2	>1/2	10	900
3	>1/2	10	900

Graph 1: **N/A**

Photo 1:



Analysis and Summary:

This testing shows the prototype is able to withstand the weight of the largest sockets that Antonio's Auto Shop has in the amount specified which means the prototype matches that testing parameter adequately

Implications and Next Steps:

The prototype is ready in this area

Test 3 – Ability to hold 10 ½ inch sockets securely and indefinitely

Requirement Priority: High

(Requirement: The prototype must be able to hold 10 ½ inch sockets on itself in the designed slots securely and indefinitely)

Testing Procedure:

Independent Variable: The socket sizes on the object

Dependent Variable: Whether the prototype breaks or not

Controls: Material object is on, sockets used, prototype remains still

Each test was repeated 1 times per socket used

All other conditions held constant

Raw Data:

Trial #	Socket size (In)	Did the proto remain unchanged
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1	1/4	Yes
2	3/8	Yes
3	1/2	Yes

Analysis and Summary:

The prototype was able to hold all 10 sockets of each size without them breaking the prototype or altering it in any meaningful way which means the test was successful

Implications and Next Steps:

The prototype matches this requirement adequately and is ready for production

